

Agency, Communion, and Proxemics as Universal Control Variables for Socially Expressive Robots

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Abstract—Robots today excel at task execution yet remain poor at being *understood* in social contexts. We propose a principled framework that parametrises robot actions within a three-dimensional social space defined by Agency, Communion, and Proxemics (ACP). We theoretically ground these variables in social psychology, map them onto robot motion control, and introduce a reinforcement learning and generative modelling pipeline to realise socially expressive behaviours. Through controlled human–robot interaction studies, we demonstrate that ACP variables systematically alter human perception of robots’ trustworthiness, politeness, and cooperation. Our findings establish ACP as a minimal universal set of control variables for socially expressive embodied AI.

Index Terms—Humanoid robots, embodied AI, reinforcement learning, motion generation, social interaction, agency, communion, proxemics.

I. INTRODUCTION

ROBOTS are increasingly deployed in human-centered environments, yet their acceptance and trust hinge not only on functional competence but on their ability to communicate social intent [1]. Humans interpret actions along latent socio-affective dimensions; however, robot control traditionally optimises only task-level goals (*e.g.*, stability, precision). This disconnect leads to socially awkward or unsafe interactions.

Our proposition: grounding robot motion generation in three fundamental social variables—*Agency* (assertiveness), *Communion* (affiliativeness), and *Proxemics* (interpersonal distance) [2], [3]. We hypothesise that ACP coordinates determine how humans perceive and cooperate with robots, and that parametrising robot control in this space can yield predictable, interpretable, and safe social behaviours.

Core novelty. Unlike affect labels or single-cue heuristics (*e.g.*, gaze), we elevate *social variables to first-class control parameters* that are optimised jointly with physical feasibility. We further show that these variables constitute a compact control basis that generalises across platforms (humanoid / manipulator-on-quadruped) and tasks (passing, handover, side-by-side walking).

II. RELATED WORK

A. Embodied AI and Social Interaction

Large language/vision-language models have recently been extended to robots, but their social grounding remains weak. *Science Robotics* has highlighted that safe social interaction requires principles of interpretability and uncertainty awareness [1].

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B. Social Dimensions in Psychology

- **Agency–Communion:** canonical axes of interpersonal behaviour [2].
- **Proxemics:** interpersonal distance zones (intimate, personal, social, public) [3].
- **Valence–Arousal–Dominance:** widely used in affective computing [4], but less actionable in control.

C. Robotics and Social Variables

Prior work has studied gaze [1], interpersonal distance, or affect recognition, but no generalisable control framework has been proposed that formalises multiple social variables as *first-class control parameters*.

blabla...

III. THEORETICAL FRAMEWORK

We formalise ACP as follows. Let $a \in [0, 1]$, $c \in [0, 1]$, and $p \in [0.5, 2.5]$ m be continuous control variables.

- **Agency** (a): mapped to motion onset latency Δt_{start} , peak velocity v_{max} , and torque allocation/actuation aggressiveness. High a = proactive, forceful; low a = hesitant, reactive.
- **Communion** (c): mapped to trajectory smoothness (low jerk), lateral yielding displacement Δy , and openness of posture (arm abduction and torso yaw). High c = affiliative, cooperative.
- **Proxemics** (p): mapped to the target interpersonal distance band (personal/social/public). Enforced as a soft constraint on robot–human separation $d(t)$ around a target band $[d^-, d^+]$ centred at p .

A. Operationalisation via Kinematic Proxies

Given a motion segment $X_{1:T}$ (skeleton or task-space markers), we compute weak labels:

$$a = \alpha_1 \left(1 - \frac{\Delta t_{\text{start}}}{T_0}\right) + \alpha_2 \text{norm}(v_{\text{max}}) + \alpha_3 \text{norm}(\bar{a}), \quad (1)$$

$$c = \beta_1 \text{norm}(\text{arm-open}) + \beta_2 (1 - \text{norm}(\text{jerk})) + \beta_3 \text{norm}(\Delta y / \Delta x), \quad (2)$$

$$p = \gamma_1 \text{norm}(d_{\text{min}}) + \gamma_2 \text{norm}(\bar{d}) + \gamma_3 \text{norm}(d_{\text{stop}}), \quad (3)$$

where $\text{norm}(\cdot)$ is per-dataset z-score then min–max scaling.

Remark: When datasets lack explicit counterparts, p can be supplied at control time while a, c are supervised using proxies only.

IV. TECHNICAL REALISATION

A. Conditional Motion Generator

We seek $G(\text{scene}, a, c, p) \rightarrow \text{traj}$, with two implementation lines.

1) *Line A: Conditional Regression (MVP)*: A temporal Transformer maps (scene, a, c, p) to a low-dimensional reference trajectory. Loss combines trajectory reconstruction, conditional alignment, and physical feasibility.

2) *Line B: Conditional Motion Diffusion*: We adopt a DiT-style denoiser with classifier-free guidance on (a, c, p) , plus conditional alignment regularisation and projected guidance for physical plausibility.

B. Socially-Constrained RL Control

Low-level controllers are trained in Isaac Lab (PPO/SAC) to track generator outputs while optimising social rewards:

$$r = w_{\text{task}} r_{\text{task}} + w_{\text{ag}} r_{\text{Agency}} + w_{\text{com}} r_{\text{Communion}} + w_{\text{prox}} r_{\text{Proxemics}}, \quad (4)$$

$$r_{\text{Agency}} = f_{\text{ag}}(\Delta t_{\text{start}}, v_{\text{max}}), \quad (5)$$

$$r_{\text{Communion}} = f_{\text{com}}(\Delta y / \Delta x, \text{jerk}^{-1}), \quad (6)$$

$$r_{\text{Proxemics}} = -\|d(t) - p\|^2. \quad (7)$$

C. Uncertainty-Aware Gating

We estimate perception uncertainty via MC-dropout on the human pose estimator. If entropy exceeds threshold, fallback policy is triggered: $(a, c, p) = (a_{\text{min}}, c_{\text{mid}}, p_{\text{public}})$.

D. Cross-Platform Deployment

We validate on humanoid and quadruped-with-arm platforms using the same (a, c, p) interface, only tuning WBC parameters.

V. EXPERIMENTS

A. Tasks

- 1) Passing in a corridor (Proxemics, Communion).
- 2) Object handover (Agency, Proxemics).
- 3) Side-by-side walking (Agency, Communion).

B. Datasets and Implementation

AMASS for base kinematics; BEHAVE and in-house MoCap for proxemics calibration. Temporal Transformer baseline; DiT diffusion upgrade. RL trained in Isaac Lab with 4096 envs.

C. Metrics

- **Objective**: separation, yielding displacement, completion time, stability, energy.
- **Behavioural**: user acceptance, avoidance, pace synchrony.
- **Subjective**: 7-point Likert on politeness, trust, cooperativeness.

D. User Study Protocol

1) *Participants*: 30 participants (balanced gender, age 20–40) recruited under institutional ethics approval.

2) *Procedure*: Each participant experiences robots performing tasks under different (a, c, p) conditions and baseline controllers. Trials are randomised.

3) *Measures*: Participants complete post-trial questionnaires (Likert scales). Behavioural data (acceptance, avoidance) are logged via motion capture.

4) *Analysis*: Mixed-effects ANOVA across conditions; Holm correction for multiple comparisons. Effect sizes reported.

E. Baselines

Task-only, Valence–Arousal generation, end-to-end LLM planner.

F. Ablations

Remove social rewards, disable generator, disable gating, swap ACP with alternative style conditions.

VI. RESULTS

ACP settings produce consistent shifts in perception and behaviour; high a , high c , social-zone p maximise cooperation. Mismatched ACP yields avoidance. Uncertainty gating reduces missteps under occlusion. Portability confirmed across platforms.

VII. DISCUSSION

A. Scientific Contribution

ACP is validated as a universal social control basis: continuous, optimisable, and platform-agnostic.

B. Limitations and Future Work

Tasks are short scripted; cultural differences not studied. Proxemics supervision partly synthetic.

C. Reproducibility and Ethics

Code and data dictionaries will be released. User study under ethics approval, informed consent obtained.

VIII. CONCLUSION

We establish Agency, Communion, and Proxemics as first-class variables in robot motion generation. By uniting generative modelling, socially-constrained RL, and uncertainty-aware gating, we demonstrate ACP governs human interpretation of robot behaviour.

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